

Vertical GaN Power Devices: Progress in Materials, Device Structures, and Breakdown-Field Management

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As the global demand for efficient energy conversion accelerates—driven by Artificial Intelligence (AI) data centers and Electric Vehicle (EV) electrification—Silicon (Si) power devices have approached their theoretical limits. Gallium Nitride (GaN) has emerged as a superior wide-bandgap (WBG) alternative. While Lateral GaN High-Electron-Mobility Transistors (HEMTs) dominate the low-voltage market, they suffer from surface-related reliability issues and poor area scaling at high voltages (>650 V). This paper presents an exhaustive review of Vertical GaN (v-GaN) technology. We analyze the fundamental physics favoring vertical architectures, including the decoupling of breakdown voltage from chip area and superior avalanche capability. Critical fabrication challenges—specifically epitaxial growth on native vs. foreign substrates, p-type doping via Mg-intercalated superlattices, and low-damage etching—are discussed in depth. We review state-of-the-art device topologies (CAVET, Trench MOSFET, FinFET) and advanced edge termination techniques (Bevel, JTE, FIT). Finally, we report on recent 2024-2025 breakthroughs, including the world's first vertical GaN-on-Si HEMT and hybrid GaN-on-SiC devices, providing a roadmap for v-GaN commercialization.

Index Terms—Vertical GaN, Wide Bandgap Semiconductors, Power MOSFET, CAVET, Edge Termination, Heteroepitaxy, Fluorine-Implanted Termination (FIT).

I. INTRODUCTION

THE demand for high-efficiency power conversion systems has driven Silicon (Si) devices to their theoretical limits. Wide Bandgap (WBG) semiconductors, particularly Gallium Nitride (GaN) and Silicon Carbide (SiC), offer significant improvements in power density and efficiency. GaN, with a critical electric field (E_c) of ≈ 3.3 MV/cm and high electron mobility, allows for devices with lower on-resistance (R_{on}) and faster switching speeds compared to Si.

Historically, the commercial GaN market has been dominated by Lateral GaN devices (GaN-on-Si HEMTs). However, as applications demand higher voltages (> 900 V) and higher power levels, the lateral architecture faces bottlenecks. Vertical GaN (V-GaN) devices, where the current flows through the bulk of

the drift layer, have recently gained significant research traction as the solution to these high-power challenges.

A. The Power Electronics Landscape

The performance and efficiency of modern power conversion systems are fundamentally constrained by the intrinsic properties of the semiconductor materials employed in their switching devices. Silicon (Si), despite its long-standing dominance in the semiconductor industry, is increasingly limited by its narrow bandgap energy of approximately 1.1 eV and relatively low critical electric field. These material constraints restrict its capability to operate reliably under high-voltage and high-temperature conditions, leading to substantial conduction and switching losses in advanced power systems.

Silicon Carbide (SiC) has emerged as a practical successor in high-voltage applications (900 V–3.3 kV),

enabled by its wider bandgap of 3.2 eV, superior critical electric field strength, and high thermal conductivity. These attributes have facilitated the commercialization of SiC MOSFETs and diodes in electric vehicles, renewable energy systems, and industrial motor drives.

Gallium Nitride (GaN), however, offers an even more favorable intrinsic material profile for next-generation power electronics. GaN exhibits a critical electric field on the order of 3.3 MV/cm—surpassing that of SiC—and an electron mobility approaching $2000 \text{ cm}^2/\text{V} \cdot \text{s}$, which is nearly three times higher than typical SiC drift-layer mobility. These combined material advantages yield a substantially enhanced Baliga Figure of Merit (BFOM), a metric that captures the fundamental trade-off between breakdown voltage (BV) and specific on-resistance ($R_{on,sp}$). When normalized to Si, the BFOM values for SiC and GaN are approximately 392 and 993, respectively, underscoring the theoretical potential of GaN for achieving simultaneously low conduction loss and high-voltage blocking capability.

In view of these intrinsic advantages, GaN is increasingly recognized as a strong candidate for high-efficiency power conversion systems, particularly where switching frequency, power density, and thermal performance are of primary concern.

B. Limitations of Lateral GaN Devices

Commercial GaN power devices are presently dominated by lateral high-electron-mobility transistors (HEMTs) fabricated on foreign substrates such as Silicon or Sapphire. While these architectures have enabled rapid market penetration in the sub-650 V regime, they exhibit several intrinsic limitations that fundamentally restrict their scalability toward higher-voltage and high-reliability applications.

A primary constraint arises from the surface-dominated electric-field distribution inherent to lateral structures. Under high drain bias, the electric field exhibits pronounced crowding at the gate–drain access region, where the peak field resides almost entirely near the device surface. As a consequence, increasing the breakdown voltage (BV) requires extending the lateral

gate–drain spacing or implementing complex field-plate structures, both of which lead to increased chip area, higher parasitic capacitances, and elevated fabrication cost. This lateral scaling requirement stands in stark contrast to vertical device architectures, in which the breakdown voltage is predominantly governed by the drift-layer thickness rather than device footprint.

Thermal management represents an additional bottleneck. Lateral GaN devices grown on foreign substrates rely on thick, low-thermal-conductivity buffer layers to mitigate lattice and thermal mismatch. These buffer stacks introduce substantial thermal resistance, impeding efficient heat extraction from the active channel. As switching frequencies and power densities increase—particularly in applications such as data-center power modules—the limited vertical heat flux in lateral GaN structures results in higher junction temperatures, degraded reliability, and constrained power-handling capability.

Device reliability is further challenged by the presence of surface states and trap-assisted charge modulation in the two-dimensional electron gas (2DEG) channel. These surface and interface traps can capture and release carriers during switching transients, leading to dynamic on-resistance (R_{on}) degradation and the well-known “current collapse” phenomenon. The susceptibility of the 2DEG to trapping effects also inhibits the ability of lateral GaN devices to exhibit avalanche behavior. Since the electric-field peak resides near the surface rather than within the bulk, lateral GaN HEMTs typically fail catastrophically during over-voltage events, limiting their suitability for applications requiring avalanche ruggedness or fault tolerance.

Collectively, these limitations underscore the fundamental architectural constraints of lateral GaN devices and motivate the exploration of vertical GaN technologies that redistribute electric fields into the drift layer, improve thermal pathways, and enable avalanche-capable high-voltage operation.

C. The Case for Vertical GaN

Vertical GaN (v-GaN) power devices fundamentally differ from their lateral counterparts in that cur-

rent is transported through the thickness of the GaN drift layer rather than along the surface. This architectural shift redistributes the electric field, modifies the breakdown characteristics, and enables significantly enhanced power-handling capability. Several intrinsic advantages arise from this vertical current-flow configuration, making v-GaN a promising candidate for high-voltage, high-density power applications.

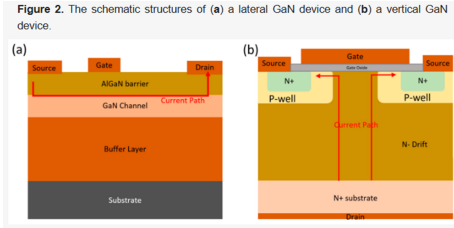


Fig. 1: lateral and vertical GaN

A principal benefit of the vertical structure is its favorable breakdown voltage (BV) scaling behavior. In contrast to lateral devices—where increasing BV necessitates enlarging the gate-drain spacing—the breakdown voltage in v-GaN is predominantly governed by the drift-layer thickness (d). To first order, the relationship follows $V_{BD} \approx E_{crit} \cdot d$, where E_{crit} denotes the material's critical electric field. This decoupling of breakdown capability from lateral device area allows vertical devices to achieve high-voltage ratings while retaining compact footprints and reduced parasitic capacitances, yielding superior area efficiency compared to lateral GaN HEMTs.

Another key advantage is the improved avalanche robustness afforded by the bulk-centric electric-field distribution. By shifting the peak electric field away from the vulnerable surface region and into the uniformly doped drift epitaxy, v-GaN devices are capable of sustaining controlled avalanche events without catastrophic failure. This capability is essential for applications where fault tolerance, surge reliability, or transient overvoltage conditions are present. Such avalanche behavior is fundamentally inaccessible to lateral GaN devices, whose surface-dominated electric-field profiles typically result in destructive breakdown.

Furthermore, vertical GaN devices can support

substantially higher current densities owing to their utilization of the full three-dimensional volume of the drift region for charge transport. The vertical conduction path reduces current crowding, enhances current spreading, and enables a larger effective cross-sectional area for conduction compared to the narrow lateral surface channel in HEMT structures. As a result, v-GaN devices achieve higher maximum current density (J_{max}) and improved thermal distribution, both of which are critical for high-power switching and compact module integration.

Collectively, these attributes—scalable breakdown voltage, inherent avalanche capability, and high current density—underscore the fundamental advantages of the vertical GaN architecture and motivate its development as a next-generation platform for high-voltage, high-efficiency power electronics.

II. MATERIAL GROWTH AND SUBSTRATE CHALLENGES

The fabrication of v-GaN devices is fundamentally limited by the availability and quality of substrates.

A. Homoepitaxy (GaN-on-GaN)

Growing GaN devices on native GaN substrates is the ideal solution. It ensures zero lattice mismatch and zero thermal expansion mismatch.

GaN-on-GaN Offers High Ruggedness

Vertical construction of GaN-on-GaN offers innate robustness based on its homoepitaxial structure, compared to lateral devices such as GaN-on-Si / GaN-on-SiC

Attribute	GaN-on-Si	GaN-on-SiC	GaN-on-GaN	Impact
Defect Density (cm^{-2})	10^6	5×10^5	10^1 to 10^2	Better Yields, Higher Reliability
Lattice Mismatch (%)	17	3.5	0	Elimination High stress in Epitaxy causing Dislocations $\rightarrow$ Better Reliability
CTE Mismatch %	54	25	0	No Warpage deformations (cracking, warping)
Layer Thickness (μm)	1-2	2-6	>40	Higher BV $\rightarrow$ Higher voltage applications

Fig. 2: GaN on GaN

- **Defect Density:** Homoepitaxy allows for dislocation densities as low as $10^3 - 10^5 \text{ cm}^{-2}$, which is critical for reducing leakage current and preventing premature breakdown.
- **Growth Methods:** Hydride Vapor Phase Epitaxy (HVPE) is the industry standard for producing thick GaN substrates, offering growth rates of 50–100 $\mu\text{m/h}$. Alternatively, the Na-flux

method (High Nitrogen Pressure Solution Growth) can produce high-quality crystals but requires expensive high-pressure equipment (up to 20 kbar).

- **Cost Barrier:** The primary drawback is cost. Native GaN substrates remain significantly more expensive than Si or SiC.

B. Heteroepitaxy (GaN-on-Foreign Substrates)

To reduce costs, researchers explore growing vertical structures on foreign substrates.

GaN-on-Sapphire: While cheap, Sapphire has a 16% lattice mismatch and poor thermal conductivity ($0.25 \text{ W/cm} \cdot \text{K}$), leading to high dislocation densities ($> 10^8 \text{ cm}^{-2}$).

GaN-on-Si: Silicon is the most cost-effective substrate but presents a -17% lattice mismatch and a significant thermal expansion coefficient mismatch (2.6 vs $5.6 \times 10^{-6} \text{ K}^{-1}$), often resulting in wafer bowing or cracking during thick film growth.

Impact on Leakage: As shown in leakage current analysis, devices on foreign substrates exhibit leakage currents orders of magnitude higher than those on native substrates, dominated by Variable-Range Hopping (VRH) through dislocation defects.

III. CRITICAL PROCESS TECHNOLOGIES

Successful v-GaN fabrication requires mastering three key processes: Epitaxy, Doping, and Etching.

A. P-Type Doping and Mg Activation

Achieving highly conductive p-GaN is the "Achilles' heel" of GaN technology. Magnesium (Mg) is the standard acceptor dopant, but it suffers from:

High Activation Energy: Mg has a deep acceptor level, meaning very few dopants ionize at room temperature.

Self-Compensation: High Mg concentrations often lead to the formation of compensating defects (e.g., nitrogen vacancies) that reduce hole concentration.

Recent Breakthrough (2024): Wang et al. demonstrated 2D-magnesium-intercalated GaN superlattices (MiGs). By annealing Mg thin films, they achieved interstitial diffusion of Mg into monolayer atomic sheets.

This technique decouples carrier mobility from concentration, enabling p-GaN with high hole density and high mobility—a critical enabler for low-resistance contacts in v-GaN devices.

B. Advanced Etching Techniques

Vertical devices like Trench MOSFETs and FinFETs require deep etching of GaN. Standard Inductively Coupled Plasma (ICP) etching using Cl_2/BCl_3 allows for anisotropic profiles but introduces plasma-induced damage (Si impurities, N-vacancies) at the sidewalls.

Hybrid Etching: A combination of ICP dry etching followed by hot Tetramethylammonium Hydroxide (TMAH) wet etching is essential. TMAH selectively removes the damaged layer, smoothing the sidewalls (reducing roughness R_q) and aligning the trench to the m-plane or a-plane for optimal channel mobility.

IV. DEVICE ARCHITECTURES

Several V-GaN topologies are currently under intense research and early commercialization.

A. Current Aperture Vertical Electron Transistor (CAVET)

The CAVET combines the high mobility of the AlGaIn/GaN channel (similar to HEMT) with a vertical drift region. Current flows through an aperture into the bulk. This structure offers low channel resistance but involves complex processing to create the current blocking layer (CBL).

The CAVET combines the high channel mobility of an AlGaIn/GaN heterojunction (like a HEMT) with a vertical drift region.

Operation: Current flows through a 2DEG channel at the surface, then passes through an "aperture" in a Current Blocking Layer (CBL) to the drain.

Advantages: It avoids the need for p-type channel doping and utilizes the polarization-induced 2DEG for low on-resistance.

Challenges: Fabricating the CBL (usually p-GaN or ion-implanted regions) requires complex regrowth steps that can introduce interface impurities.

B. Trench MOSFETs

Trench gate structures are standard in Si and SiC technology. In GaN, etching the trench without creating damaging defects is challenging. However, recent progress in low-damage etching and dielectric deposition has led to V-GaN Trench MOSFETs with stable threshold voltages ($V_{th} > 3V$) and low specific on-resistance.

Trench MOSFETs are the most promising topology for normally-off operation.

Structure: A gate trench is etched through the n+ source and p-body layers into the drift region. A channel is induced along the trench sidewall.

Performance: Recent devices have demonstrated breakdown voltages >1200 V. The key metric is minimizing the specific on-resistance ($R_{on,sp}$) while maintaining a stable threshold voltage (V_{th}).

Reliability: The gate oxide dielectric quality is critical. High electric fields at the trench corners can cause dielectric breakdown. Rounding the trench corners via wet etching helps mitigate this.

C. Vertical FinFETs

FinFETs utilize narrow semiconductor "fins" wrapped by the gate. This provides superior electrostatic control, suppressing short-channel effects and lowering leakage current. Recent demonstrations include optically triggered FinFETs, which isolate the gate driver from high-voltage noise.

V. EDGE TERMINATION TECHNOLOGIES

In vertical power devices, the electric field tends to crowd at the edge of the active area, leading to premature breakdown. Effective edge termination is required to spread this field and approach the theoretical breakdown limit of the material.

A. Bevel Mesa Termination

This involves etching the device edge at a specific angle (θ).

Positive Bevel: The cross-sectional area decreases from the heavily doped to lightly doped region.

Negative Bevel: The area increases.

Physics: A shallow bevel angle stretches the depletion region along the surface, reducing the peak surface electric field (E_{peak}). However, achieving precise angles ($< 10^\circ$) is challenging in manufacturing.

B. Junction Termination Extension (JTE)

JTE involves creating zones of varying p-type charge (P_1, P_2, P_3) at the edge. By grading the charge, the electric field is gradually reduced³⁵. Simulation results show that optimizing the JTE dose and length can effectively flatten the electric field profile, preventing crowding.

C. Guard Rings (GR)

Floating p-type rings are placed around the main junction. When the depletion region reaches a ring, it "punches through," allowing the ring to share the potential drop. Guard rings are highly robust but require precise spacing design.

D. Fluorine-Implanted Termination (FIT) - 2025 Breakthrough

A collaboration between Huawei and Shandong University introduced the FIT-MOS.

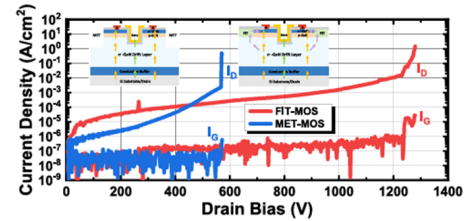


Fig. 3: FIT

Mechanism: Fluorine ions implanted into GaN act as fixed negative charges. This artificially expands the depletion region and shields the high-field areas without the need for complex p-GaN regrowth.

Result: The FIT-MOS achieved a breakdown voltage of 1277 V, compared to only 567 V for a standard mesa-etched termination (MET) device³⁹. The device maintained a low $R_{on,sp}$ of $5.6 \text{ m}\Omega \cdot \text{cm}^2$.

VI. RECENT BREAKTHROUGHS (2024-2025)

The period of 2024-2025 has seen transformative advances in v-GaN, pushing it closer to commercialization.

A. World's First Vertical GaN HEMT on Silicon (Ziener)

Dr. Li Zilan and the team at Guangdong Ziener achieved a milestone by fabricating a vertical GaN HEMT on a Silicon substrate.

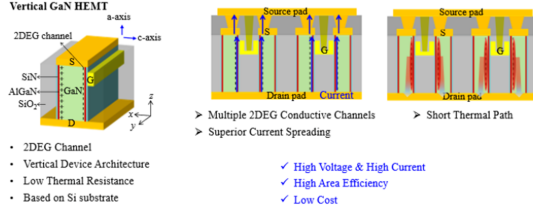


Fig. 4: vGaN HEMT

Innovation: They successfully epitaxially grew a vertical 2DEG channel structure directly on Si.

Significance: This architecture combines the low cost and large wafer size (8-inch) of Silicon with the vertical current conduction benefits of v-GaN (high current spreading, short thermal path).

Impact: This technology disrupts the cost structure, potentially making v-GaN competitive with Si-IGBTs and lateral GaN in cost-sensitive markets.

B. Hybrid GaN-on-SiC Devices

Researchers have developed hybrid devices that use a SiC substrate as the drain and drift region, with a GaN layer on top for the channel.

Synergy: This structure leverages SiC's superior thermal conductivity (3x that of Si, 1.5x that of GaN) and avalanche capability, while utilizing GaN's high electron mobility for fast switching.

Performance: Hybrid GaN-on-SiC trench MOSFETs have demonstrated fully vertical conduction with reduced thermal resistance compared to GaN-on-Si.

C. Optically Triggered Power Devices

MIT researchers demonstrated optically controlled vertical GaN FinFETs. By integrating a photosensitive gate, the device can be switched using light. This eliminates the need for electrical gate drivers and galvanic isolation components, significantly reducing the size and EMI susceptibility of high-voltage power modules.

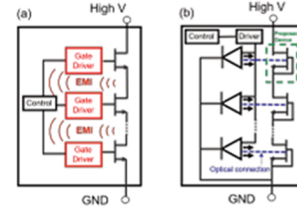


Fig. 5: Optical Trigger

VII. APPLICATIONS AND FUTURE OUTLOOK

The unique properties of v-GaN position it for specific high-power sectors where lateral GaN and SiC face limitations.

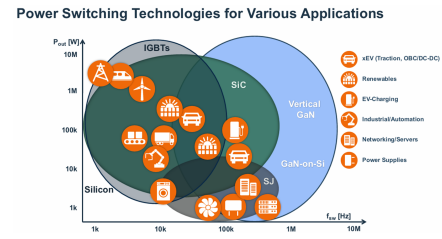


Fig. 6: Applications

A. AI Data Centers

AI processors require massive currents at low voltages (48V to <1V conversion). Vertical GaN devices allow for 3x higher power density in DC-DC converters. By enabling higher switching frequencies, v-GaN reduces the size of passive components (inductors/capacitors), freeing up valuable board space for compute engines.

B. Electric Vehicle (EV) Powertrains

For 800V battery architectures, v-GaN offers an ideal solution for On-Board Chargers (OBC) and traction inverters. ONSEMI has announced 1200V v-GaN devices sampling in 2025, targeting mass production by 2026. The vertical architecture provides the necessary avalanche ruggedness required for automotive safety standards.

C. Renewable Energy Grids

Solar inverters and wind turbines require high-voltage switching. v-GaN devices rated for 1200V–3300V can replace bulky Si-IGBT modules, reducing switching losses and improving system efficiency.

VIII. CONCLUSION

Vertical GaN (v-GaN) power devices have progressed rapidly from conceptual studies to experimentally validated prototypes, marking a significant inflection point in the evolution of wide-bandgap power electronics. By enabling current transport through the bulk drift region rather than along a surface-confined 2DEG channel, vertical architectures fundamentally resolve the intrinsic scaling, electric-field crowding, and reliability limitations encountered in lateral GaN HEMTs. This shift in device topology not only permits breakdown voltage to scale with epitaxial thickness but also facilitates avalanche-capable operation and substantially higher current density, positioning v-GaN as a compelling candidate for high-voltage and high-power applications.

Despite the clear material advantages of GaN, the widespread adoption of vertical devices has historically been constrained by the cost and availability of high-quality native GaN substrates. Recent developments—including the demonstration of vertical GaN-on-Si HEMTs and GaN-on-SiC hybrid structures—provide practical pathways toward reducing substrate cost while maintaining acceptable crystalline quality and thermal performance. In parallel, significant advances in enabling process technologies such as thick-film homoepitaxy, high-activation-energy p-type doping strategies, and low-damage trench etching have expanded the design space for high-performance vertical MOSFETs, CAVETs, and FinFETs.

Edge-termination engineering has similarly advanced, with techniques such as fluorine-ion-implanted termination (FIT), optimized bevel structures, and multi-ring junction extensions enabling devices to approach the theoretical breakdown limits of GaN. Several demonstrations have already achieved voltage ratings exceeding 1.2 kV with competitive specific on-resistance, validating the scalability and robustness of v-GaN architectures.

Looking forward, continued progress in substrate engineering, defect reduction, and manufacturable process integration will be essential to transition v-GaN

from laboratory-scale prototypes to high-volume commercial deployment. With these advancements, vertical GaN is well-positioned to become a foundational technology for next-generation power electronics in data-center infrastructure, electric vehicle powertrains, renewable-energy conversion, and other high-efficiency, high-power-density applications.

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